

Introduction: Deep neural networks (DNNs) are a type of machine learning model that mimic sensory processing in the human brain through their layered computational structure. Research has been conducted on the performance of DNN models on visual processing, showing that the models develop separate systems similar to those found in the brain when optimizing for a specific task. Comparatively less research has been done with audio processing, especially with newer self-supervised learning (SSL) methods, where the model observes patterns from unlabeled data. The benefit of using SSL models over supervised learning frameworks is that significantly less data and processing power is required to produce an effective model, meaning the DNN's performance on sensory processing tasks could approach human performance. While DNNs are not literal models of the brain, understanding why they process audio the way they do may help us identify computational patterns that are also present in biological perception. In particular, this project explores how self-supervised or supervised learning methods affect the performance of DNN models on music and speech classification.

Background: Neural networks are composed of layers that process complex input data into a simple output, such as a word identified from an audio file. A DNN is composed of an input layer, one or more hidden layers, and an output layer. Each layer receives input from the previous layer, performs computations, and passes results to the next layer. Studying models across their layers provides insight into how hierarchical structures in the brain function. One such hypothesis is that middle DNN layers predict primary auditory cortical responses while later layers predict non-primary auditory cortical responses (Kell and McDermott).

Task specialization refers to how different layers of a neural network become tuned to specific tasks, such as identifying speech or music. Previous research in visual processing has shown that the brain organizes itself into task-specialized regions. For example, DNNs trained on both facial and object recognition naturally develop separate systems for each task that mirror fMRI scans of humans performing the same task, suggesting that the brain also develops specialized systems (Dobs et al.). A similar effect has been observed in auditory processing, where a DNN trained on speech and genre classification tasks could predict auditory cortical fMRI responses, suggesting that the auditory cortex is hierarchically divided into a primary and non-primary region (Kell et al.). However, using a supervised learning approach to training DNNs requires a considerable amount of pre-labeled data, and brains do not naturally learn from labeled datasets.

Instead, using SSL approaches can achieve similar results without requiring labeled data. Recent work with SSL models has shown their ability to predict hierarchical auditory cortical responses to speech (Vaidya et al.). One study trained an SSL model, Wav2Vec 2.0, on fMRI data of people listening to speech and found that the model performed similarly to humans while also revealing task-specific optimization (Millet et al.).

Furthermore, using cochleagrams as input data to the model and adding noise to them can also imitate biological environments (Tuckute et al.). Cochleagrams are time-frequency representations of audio that mimic the human ear. Adding background noise, or superimposing other audio onto the cochleagrams, can improve the classification accuracy of the DNN if it is trained on multiple tasks. Few studies have been conducted with the combination of self-supervised learning and cochleagram input data, which this project will focus on.

Achieving a better understanding of auditory processing has important applications. For example, one model imitating the human ear and head structure reproduced human-like sound localization, which can improve spatial awareness in hearing aid and virtual reality algorithms (Francl and McDermott). Studying why DNNs specialize in specific tasks may offer insights into which features of sound matter most in auditory processing, leading to more biologically grounded models of brain function (Kanwisher et al.).

Therefore, my research question is: **how do different supervised learning objectives affect auditory specialization in DNN models trained on speech and music?**

Methodology: To address my research question, I will compare an SSL model to a supervised learning model and compare their performance on speech and music genre classification tasks. My approach will involve adjusting the parameters of an SSL model and analyzing how its performance changes at each layer.

I will use two publicly available neural networks: VGGish for supervised learning (Hershey et al.), and Wav2Vec 2.0 for self-supervised learning (Baevski et al.). Both models are built in Python and are designed for audio-processing tasks. Since DNNs require large computational resources, I will connect a standalone GPU (graphics processing unit) to my computer to accelerate training. Unlike CPU (central processing unit) usage, which is relatively slow for deep-learning applications, a GPU setup will enable faster iterations through the training data. I will also purchase an SSD (solid-state drive) where I will install the training data, it will have significantly faster file read/write speeds compared to the hard-disk drive my computer currently has.

I will use data that exists for the CONAP project in the Peer Herholz lab (Herholz et al., under review). This dataset converts large publicly available music and audiobook datasets into cochleagrams. Before training, I will preprocess the data to ensure compatibility with the models. This preprocessing will involve adding environment noise to the cochleagrams and dividing the speech files into 3-5 second chunks for word classification.

In order to train the models, I will prepare them to take cochleagram data as an input. I will train both models on word and genre classification tasks where they will need to specify a word being spoken or the genre of a piece of music. For VGGish, I will measure its accuracy along a confusion matrix to see where the model is making classification errors since it is common for false negatives or false positives to appear. I would spend more time with Wav2Vec 2.0 where I would adjust the learning functions, which change how layers in the neural network operate. I would use the cross-entropy loss and mean squared error learning functions, however I may choose to experiment with others if there is time.

In addition to evaluating overall classification accuracy, I will conduct a layer-wide analysis of each model to see at which layer they specialize into speech or genre classification tasks. In order to visualize transformations to the data across layers, I will use Principal Component Analysis (PCA): a technique that simplifies the complex data produced by the DNN layers into a smaller number of key features. Performing PCA on those layers makes it easier to understand at which layer the model transitions from general audio features to speech-related features or genre classification, which could show how each layer contributes to the final output.

Once I have my results, I will use the Analysis of Variance (ANOVA) statistical test to compare the models across 3 conditions: supervised learning, self-supervised learning, and differing learning functions. To look for any significant difference in performance between the models at certain layers, I will look for p-values < 0.05 in accuracy. I will then use R to create data visualizations of statistically significant results, as well as examining the reasons why some are not statistically significant.

Qualifications: In this project, I will need to implement machine learning onto large datasets and analyze the models' performance. I am currently training in Python and machine learning through the Peer Herholz lab through the Undergraduate Research Assistant Program where I have been working 11 hours a week (Winter and Spring 2025). Furthermore, I have taken two introductory programming courses at Northwestern: CS 110 as an introduction to Python, and STAT 202 as an introduction to statistics and data science. In data science, I learned about data wrangling and multiple linear regression which will help with my analysis of the results. I also have a background understanding of the auditory cortex as I took NEUROSCI 107, a seminar in the basics of neuroscience and literature review. I aim to build on my knowledge of computational neuroscience so I can learn to develop AI models and contribute to advancements in neurotechnology. In the future, I hope to apply my learning to fields related to cognitive science or environmental science.

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